

HATTULA LEPAA AWS, Finland

WMO: 027040

Lat: 61.115N

Lon: 24.324E

Elev: 87

StdP: 100.28

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.6	-20.0	-26.1	0.4	-23.3	-22.4	0.5	-19.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	27.7	18.8	25.7	17.9	23.8	17.1	20.5	25.0	19.2	23.8	18.0	22.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.9	22.5	17.5	12.7	21.2	16.3	11.7	19.8	59.4	25.3	55.1	23.9	51.2	22.5	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-24.5	29.5	5.9	2.3	-28.8	31.1	-32.3	32.4	-35.6	33.7	-39.9	35.4		
(5)			-24.7	21.6	5.9	2.1	-29.0	23.1	-32.4	24.3	-35.8	25.5	-40.0	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-6.5	-5.3	-2.0	4.1	10.8	14.5	17.7	15.9	11.2	4.8	1.2	-2.9		
	DBStd	9.31	6.15	6.52	4.70	3.60	4.20	3.43	3.24	2.80	3.17	3.47	4.10	5.67		
	HDD10.0	2389	513	428	371	179	41	4	0	0	24	164	264	401		
	HDD18.3	4791	771	662	629	426	234	123	53	86	213	420	514	659		
	CDD10.0	692	0	0	0	3	66	139	237	182	61	2	0	0		
	CDD18.3	51	0	0	0	0	2	8	32	9	0	0	0	0		
	CDH23.3	505	0	0	0	0	41	102	290	73	0	0	0	0		
	CDH26.7	99	0	0	0	0	3	17	68	12	0	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	590	47	34	31	30	38	64	72	68	47	64	50	46		
	PrecMax	678	78	74	57	72	80	111	166	134	139	161	106	96		
	PrecMin	443	7	2	4	3	9	29	4	12	10	19	4	6		
	PrecStd	60	18	20	13	18	20	23	44	33	34	31	26	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	5.4	10.0	20.3	26.9	28.8	30.7	28.4	21.8	14.0	10.1	8.2		
		MCWB	3.0	2.9	4.9	11.1	15.5	18.6	20.1	19.4	16.8	12.0	9.1	7.6		
	2%	DB	2.6	3.5	7.1	16.3	24.2	26.2	28.6	25.2	19.3	12.2	8.9	5.5		
		MCWB	2.1	2.4	3.8	8.7	14.8	17.3	19.9	18.2	15.2	10.8	8.4	4.8		
	5%	DB	1.6	2.5	5.1	13.6	21.9	24.0	26.6	23.3	17.6	11.2	7.3	4.1		
		MCWB	1.2	1.8	2.6	7.7	14.1	16.5	19.1	17.2	14.2	10.2	6.8	3.5		
	10%	DB	0.8	1.6	3.6	10.9	19.4	21.7	24.5	21.5	16.2	10.1	6.2	2.9		
		MCWB	0.4	1.0	1.8	5.9	12.4	15.3	18.4	16.2	13.9	9.0	5.7	2.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	3.4	5.8	11.9	17.9	20.2	22.4	21.3	17.3	12.6	9.5	7.7		
		MCDB	3.5	4.3	8.7	18.0	23.4	25.1	27.5	25.3	20.4	13.6	10.0	8.1		
	2%	WB	2.2	2.6	4.2	9.9	16.2	18.6	21.2	19.2	16.0	11.4	8.2	4.8		
		MCDB	2.5	3.3	6.3	15.8	22.1	24.4	26.0	23.0	18.4	11.9	8.8	5.3		
	5%	WB	1.2	1.9	3.1	8.0	14.8	17.2	20.1	18.1	15.0	10.4	6.9	3.5		
		MCDB	1.4	2.5	4.6	12.3	19.8	22.3	24.7	21.7	16.9	11.0	7.3	3.9		
	10%	WB	0.4	1.1	2.1	6.6	13.1	16.0	18.9	17.0	14.0	9.1	5.8	2.5		
		MCDB	0.7	1.6	3.4	10.1	18.7	20.7	23.5	20.3	15.7	9.9	6.1	2.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	5.7	7.8	9.5	10.9	10.0	9.9	9.4	7.5	5.6	3.6	4.2		
		MCDBR	3.3	4.7	8.4	14.8	15.4	14.1	12.9	12.4	9.8	6.5	4.3	4.4		
	5% WB	MCWBR	3.3	4.0	5.5	8.1	7.3	6.6	5.5	6.1	6.1	5.0	4.2	4.3		
		MCWBR	3.3	4.4	7.6	13.1	13.0	12.4	11.5	10.6	8.7	5.6	4.2	4.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.256	0.297	0.325	0.356	0.353	0.350	0.373	0.362	0.336	0.323	0.281	0.368		
		taud	2.091	2.165	2.246	2.375	2.440	2.472	2.424	2.449	2.515	2.408	2.192	2.616		
	Ebn at Noon	Ebn at Noon	459	669	788	839	871	879	845	826	787	652	430	312		
		Edh at Noon	40	70	92	98	100	98	101	91	72	57	36	25		
(44) All-Sky Solar Radiation	RadAvg	0.23	0.80	2.27	3.74	5.06	5.30	5.10	3.96	2.35	1.01	0.27	0.13			
	RadStd	0.04	0.18	0.29	0.39	0.55	0.42	0.50	0.44	0.31	0.18	0.05	0.02			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (60 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	-140	N/A	N/A	N/A

Nomenclature: See separate page